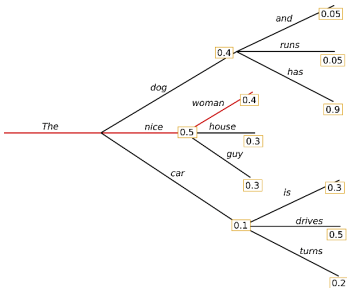


# Deep Learning for NLP

## Decoding Strategies

## Stochastic Decoding & CS/CD



### Learning goals

- Get to know different stochastic decoding strategies
- Learn about sampling with temperature, top-k sampling and top-p (nucleus) sampling
- learn about contrastive search and contrastive decoding

# SAMPLING MOTIVATION I

- *Creativity and Variation*: Sampling methods produce varied outputs for the same input, useful in creative applications like story generation and dialogue systems.
- *Avoiding Repetition*: These methods are less likely to generate repetitive loops compared to deterministic methods.

# SAMPLING (WITH TEMPERATURE) (1) I

The next token is selected randomly based on its conditional probability distribution. To control the randomness of the output sequence, a temperature parameter can be applied to the softmax function

$$\sigma(z_i) = \frac{e^{\frac{z_i}{temp}}}{\sum_{j=1}^N e^{\frac{z_j}{temp}}}$$

- $temp \rightarrow \infty$  : Output distribution  $\approx$  Uniform distribution
- $temp \rightarrow 0$  : Output distribution  $\approx$  Point mass (Greedy search)

# SAMPLING (WITH TEMPERATURE) (2) I

**Prompt: "Once upon a time"**

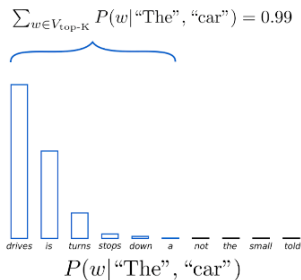
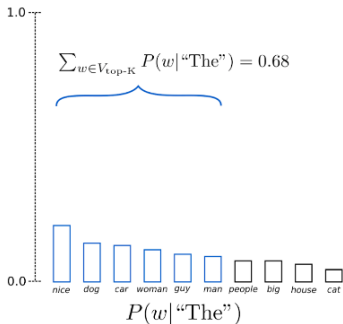
- Sampling with low temperature: *", during the Second World War, during the final months for his three most talented young players, the coach, Harry Gregg said this"*
- Sampling with high temperature: *"— well. Nowhere you call back my call, not on time; never the two on account my four. Do not come." This old woman — you might have liked, she herself — she did smile."*

The generated stories are diverse but sometimes very erratic.

⇒ Sample from the top- $k$  tokens

# TOP-K SAMPLING ► Fan et al., 2018

In Top- $k$  sampling, the  $k$  most likely next tokens are filtered, and the probability mass is redistributed. Visualization for  $k = 6$  in two sampling steps:



► huggingface, Patrick von Platen

# TOP-K SAMPLING I

**Prompt: "Once upon a time"**

- Top- $k$  ,  $k = 100$ : *"when I was young the internet was a mysterious landscape full of new and exciting ideas. I read ebooks, watched videos, read short stories"*

The quality has improved, but the fixed  $k$  might be counterproductive

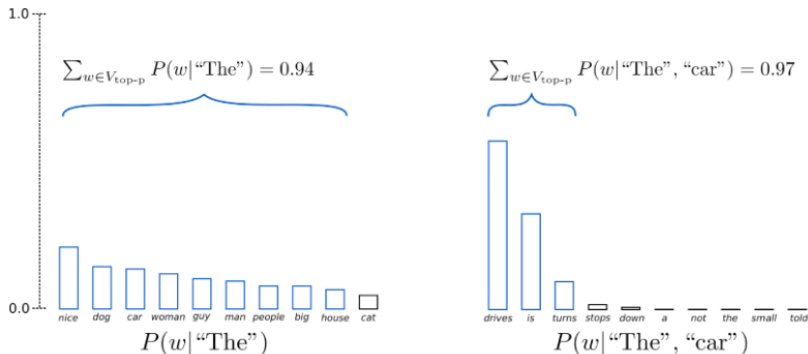
⇒ Make  $k$  dynamic

# TOP- $P$ (NUCLEUS) SAMPLING

► Holtzman et al., 2019

I

Top- $p$  sampling chooses from the smallest possible set of tokens whose cumulative probability exceeds the probability threshold  $p$ . The probability mass is then redistributed accordingly. Visualization with a threshold  $p = 0.92$ :



► huggingface, Patrick von Platen

# TOP- $P$ (NUCLEUS) SAMPLING I

## Prompt: "Once upon a time"

- Top- $p$  ,  $p = 0.92$ : *"there were four major political parties in the United States. Since then, however, they have become even more of a novelty. For the past few decades, there have been only two."*

SOTA for many years, default decoding strategy in various GPT versions, but sometimes erratic depending on  $p$  and the sampled tokens.

**Question:** Can there be a balance of coherence and diversity?

⇒ Contrastive search



$$x_t = \arg \max_{v \in V^{(k)}} \left\{ (1 - \alpha) \times \underbrace{p_{\theta}(v | \mathbf{x}_{<t})}_{\text{model confidence}} - \alpha \times \underbrace{(\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\})}_{\text{degeneration penalty}} \right\}$$

When generating output, contrastive search jointly considers:

- The probability predicted by the language model to maintain the semantic coherence between the generated text and the prompt.
- The similarity with respect to the previous context to avoid degeneration (as in Greedy or Beam search)

⇒ An "ideal" token should have a high probability and bring diversity to the story.

Empirical studies suggest  $k \in \{5, 8, 10, 15\}$  and  $\alpha \in \{0.4, 0.5, 0.6\}$

► Su & Collier, 2023

► Su & Xu, 2022

► Su et al., 2022

# CONTRASTIVE SEARCH FORMULA I

► huggingface, Tian Lan

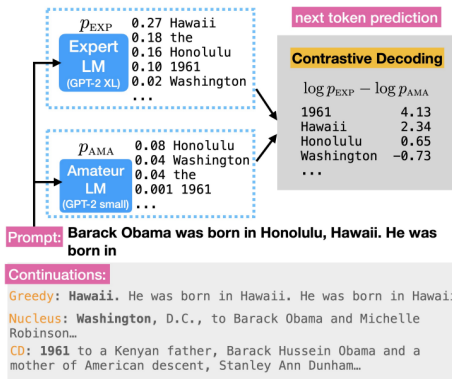
*Let's have a closer look at the formula for Contrastive Search:*

$$x_t = \underset{v \in V^{(k)}}{\operatorname{argmax}} \left\{ (1 - \alpha) \times p_{\theta}(v | \mathbf{x}_{<t}) - \alpha \times (\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\}) \right\}$$

- $x_t$  is the output token and  $\mathbf{x}_{<t}$  the context
- $V^{(k)}$  is the set of top-k predictions from the models probability distribution (this is the same  $k$  as in the top-k sampling from earlier)
- $p_{\theta}(v | \mathbf{x}_{<t})$ , the *model confidence*, is the probability of a candidate token  $v$  given the context
- $\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\}$ , the *degeneration penalty*, measures how similar  $v$  is to the context,  $s()$  is the cosine similarity between the token representations

# CONTRASTIVE SEARCH FORMULA II

- The degeneration penalty is defined as the maximum cosine similarity between the token representation of  $v$ , i.e  $h_v$ , and of all tokens in the context  $\mathbf{x}_{<t}$
- $h_v$  is computed by the language model given the concatenation of  $v$  and  $\mathbf{x}_{<t}$
- In order to maximize the formula we want  $v$  to have a high probability and a low degeneration penalty
- Intuitively, a larger degeneration penalty of  $v$  means it is more similar (in the representation space) to the context, therefore more likely leading to the problem of model degeneration
- $\alpha$  determines how much weight to give to each component
- For  $\alpha = 0$  we only consider the probability and contrastive search becomes greedy search



- Contrastive decoding exploits the contrasts between expert and amateur LM of different sizes by choosing tokens that maximize their log-likelihood difference (read the paper if you are interested, not going into more detail here!)